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SETUP AND CONFIGURATION OF A LAN USING A SWITCH AND ETHERNET CABLES

AIM:

To set up and configure a Local Area Network (LAN) using a switch and Ethernet cables, assign IP addresses to the connected computers, verify connectivity using the ping command, and share files and folders across the network.

ALGORITHM:

1. Collect the required components, including four computers, a network switch, and Ethernet cables.
2. Connect all four computers to the switch using Ethernet cables.
3. Switch on the computers and ensure that the network connections are active.
4. Open Network Settings on each computer and configure the IPv4 properties.
5. Assign IP addresses to the computers in the same network.
6. Set the subnet mask to 255.0.0.0 for all the computers.
7. Save the IP configuration settings.
8. Open Command Prompt on each computer.
9. Use the ping command to check connectivity between the connected computers.
10. Select a folder, open its Properties, and enable sharing over the network.
11. Access the shared folder from another computer connected to the same LAN.
12. Verify that the computers can communicate and share files successfully.

IP ADDRESS CONFIGURATION:

PC1 – 10.1.1.1

PC2 – 10.1.1.2

PC3 – 10.1.1.3

PC4 – 10.1.1.4

Subnet Mask – 255.0.0.0

Switch Management IP Address – 10.1.1.5 (if supported and required)

COMMAND:

ping 10.1.1.2

SAMPLE OUTPUT:

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time=2ms TTL=128

Reply from 10.1.1.2: bytes=32 time=1ms TTL=128

Reply from 10.1.1.2: bytes=32 time=2ms TTL=128

Reply from 10.1.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 10.1.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

OUTPUT:

The four computers were connected to the switch using Ethernet cables. IP addresses were assigned to all computers, and connectivity was verified using the ping command. A shared folder was configured and accessed from another computer on the LAN.

RESULT:

Thus, a Local Area Network was successfully configured using a switch and Ethernet cables. IP addresses were assigned, connectivity was tested using the ping command, and file-sharing functionality between computers on the network was studied.